

LAPORAN PRAKTIKUM
WORKSHOP SISTEM INFORMASI WEB SERVER
(TIF330805)
SEMESTER III



Acara 8

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1. Tugas

- Implementasikan Database class singleton dengan PDO.
- Buat Repository atau method di Model untuk all, find, create, update, delete.
- Lengkapi Controller + View untuk CRUD Mahasiswa.
- Ulangi untuk Prodi dan Mata Kuliah.
- Pastikan JOIN digunakan saat menampilkan daftar mahasiswa (tampilkan nama prodi).
- Tambahkan konfirmasi JavaScript sebelum hapus.

2. Tugas Mandiri

- Tambahkan fitur pencarian (search) pada halaman index mahasiswa berdasarkan nama atau NIM, menggunakan LIKE dengan prepared statement.

3. Pengerjaan

- Class database dengan PDO

```
use PDO;

7 references | 0 implementations
class Database
{
    3 references
    private static ?self $instance = null;
    2 references
    private PDO $connection;

    1 reference
    private function __construct()
    {
        $config = require __DIR__ . '/../../config/database.php';
        $dsn = sprintf(
            'mysql:host=%s;dbname=%s;charset=%s',
            $config['host'],
            $config['dbname'],
            $config['charset']
        );
    }
}
```

- Method untuk CRUD

```
7 references | 2 overrides
public function all(): array
{
    return $this->db->query('SELECT * FROM ' . $this->quotedTable() . ' ORDER BY id')->fetchAll();
}

3 references | 0 overrides
public function find(int $id): ?array
{
    $statement = $this->db->prepare('SELECT * FROM ' . $this->quotedTable() . ' WHERE id = :id');
    $statement->execute(['id' => $id]);

    return $statement->fetch() ?: null;
}
```

```
3 references | 0 overrides
public function create(array $data): bool
{
    $columns = $this->quotedColumns($data);
    $statement = $this->db->prepare(
        'INSERT INTO ' . $this->quotedTable() . ' (' . implode(', ', $columns) . ') VALUES (' .
        implode(', ', array_fill(0, count($columns), '?')) . ')'
    );

    return $statement->execute(array_values($data));
}

3 references | 0 overrides
public function update(int $id, array $data): bool
{
    $columns = $this->quotedColumns($data);
    $assignments = array_map(static fn (string $column): string => $column . ' = ?', $columns);
    $statement = $this->db->prepare(
        'UPDATE ' . $this->quotedTable() . ' SET ' . implode(', ', $assignments) . ' WHERE id = ?'
    );
    $values = array_values($data);
    $values[] = $id;

    return $statement->execute($values);
}
```

```
3 references | 0 overrides
public function delete(int $id): bool
{
    $statement = $this->db->prepare('DELETE FROM ' . $this->quotedTable() . ' WHERE id = :id');

    return $statement->execute(['id' => $id]);
}
```

- Controller + View untuk CRUD

- View

```
<tbody>
  <?php foreach ($mahasiswaList as $mhs): ?>
    <tr>
      <td><?= htmlspecialchars($mhs['nim']) ?></td>
      <td><?= htmlspecialchars($mhs['nama']) ?></td>
      <td><?= htmlspecialchars($mhs['prodi'] ?? '') ?></td>
      <td><?= htmlspecialchars((string) $mhs['angkatan']) ?></td>
      <td>
        <div class="d-flex flex-wrap gap-1">
          <a href="<?= htmlspecialchars(BASE_URL) ?>/mahasiswa/<?= (int) $mhs['id'] ?>" class="btn btn-sm btn-info">Detail</a>
          <a href="<?= htmlspecialchars(BASE_URL) ?>/mahasiswa/<?= (int) $mhs['id'] ?>/edit" class="btn btn-sm btn-warning">Edit</a>
          <form action="<?= htmlspecialchars(BASE_URL) ?>/mahasiswa/<?= (int) $mhs['id'] ?>/delete" method="POST" onsubmit="return confirm('Hapus data mahasiswa ini?')">
            <button type="submit" class="btn btn-sm btn-danger">Hapus</button>
          </form>
        </div>
      </td>
    </tr>
  <?php endforeach; ?>
  <?php if ($mahasiswaList === []): ?>
    <tr><td colspan="5" class="text-center text-muted">Data mahasiswa tidak ditemukan.</td></tr>
  <?php endif; ?>
</tbody>
```

- Controller mahasiswa

```
0 references | 0 implementations
class MahasiswaController
{
    0 references | 0 overrides
    public function index(): void
    {
        $search = trim($_GET['q'] ?? '');
        $mahasiswaList = (new MahasiswaModel())->all($search);
        $content = __DIR__ . '/../Views/mahasiswa/index.php';
        require __DIR__ . '/../Views/layouts/main.php';
    }
}
```

- Controller mata kuliah

```
0 references | 0 implementations
class MatakuliahController
{
    0 references | 0 overrides
    public function index(): void
    {
        $matakuliahList = (new MatakuliahModel())->all();
        $content = __DIR__ . '/../Views/matakuliah/index.php';
        require __DIR__ . '/../Views/layouts/main.php';
    }
}
```

- Controller prodi

```
0 references | 0 implementations
class ProdiController
{
    0 references | 0 overrides
    public function index(): void
    {
        $prodiList = (new ProdiModel())->all();
        $content = __DIR__ . '/../Views/prodi/index.php';
        require __DIR__ . '/../Views/layouts/main.php';
    }

    0 references | 0 overrides
    public function create(): void
    {
        $prodi = [];
        $isEditing = false;
        $content = __DIR__ . '/../Views/prodi/create.php';
        require __DIR__ . '/../Views/layouts/main.php';
    }
}
```

- Menampilkan data mahasiswa dengan JOIN

```
7 references | 0 overrides | prototype
public function all(string $search = ''): array
{
    $sql = 'SELECT mahasiswa.id, mahasiswa.nim, mahasiswa.nama, mahasiswa.angkatan, prodi.nama
    AS prodi
    FROM mahasiswa
    LEFT JOIN prodi ON prodi.id = mahasiswa.prodi_id
    WHERE mahasiswa.nama LIKE :name OR mahasiswa.nim LIKE :nim
    ORDER BY mahasiswa.id';
    $statement = $this->db->prepare($sql);
    $term = '%' . $search . '%';
    $statement->execute(['name' => $term, 'nim' => $term]);

    return $statement->fetchAll();
}
```

- Konfirmasi Javascript

```
<form action="<?= htmlspecialchars(BASE_URL) ?>/mahasiswa/<?= (int) $mhs['id'] ?>/
delete" method="POST" onsubmit="return confirm('Hapus data mahasiswa ini?')">
    <button type="submit" class="btn btn-sm btn-danger">Hapus</button>
</form>
```

- Fitur search menggunakan LIKE

```
WHERE mahasiswa.nama LIKE :name OR mahasiswa.nim LIKE :nim
ORDER BY mahasiswa.id';
```